



South African Reserve Bank
From the Office of
the Registrar of Banks

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To banks, branches of foreign institutions, controlling companies and auditors of banks or controlling companies

Directive 2/2014 issued in terms of section 6(6) of the Banks Act, 1990 (Act No. 94 of 1990)

Matters related to changes to internal rating systems used to calculate the minimum required capital for credit risk

Executive summary

The use of internal rating systems, including statistical models and mechanical methods, for the calculation of banks' and controlling companies' (hereinafter referred to as 'banks') minimum required capital for credit risk is subject to the prior written approval of the Registrar of Banks (the Registrar). In turn, such approval imposes specific duties on the Office of the Registrar (this Office), including the duty to:

- develop a set of review procedures for ensuring that banks' systems and controls are adequate to serve as the basis for capital calculations; and
- focus on compliance with minimum regulatory requirements as a means of ensuring the overall integrity of a bank's ability to provide prudential inputs to the capital calculator.

Among other things, the Regulations relating to Banks (the Regulations) set out the minimum capital requirements for banks that have adopted the internal ratings-based (IRB) approach (hereinafter referred to as 'IRB banks') for the measurement of their exposure to credit risk, specifically with respect to rating systems used for the calculation of minimum required capital.

To enable this Office to discharge its supervisory responsibilities effectively with respect to the aforesaid rating systems, this directive contains revised requirements in terms of the approval of material changes to IRB banks' credit risk rating systems.

This directive replaces Directive 6/2013 dated 23 May 2013.

1. Introduction

- 1.1 The approval originally granted to banks to use the IRB approach to calculate the minimum required capital for credit risk relates only to those internal rating systems, including statistical models and mechanical methods, included in the original applications submitted by banks. Any material rating system changes or developments that fall outside the scope of the approval granted by this Office are subject to a formal approval process prior to being implemented.
- 1.2 This Office issued Directive 6/2013 concerning matters related to changes to rating systems used for the calculation of minimum required capital for credit risk. Based on areas of improvement identified during the review process followed by this Office and feedback received from IRB banks, this Office decided to refine the requirements of Directive 6/2013 in order to discharge its supervisory responsibilities more effectively.
- 1.3 This directive therefore serves to inform banks of these revised requirements, specifically with regard to cases where the prior written approval of this Office will be required before a bank may implement changes to its credit risk rating system.

2. Directive

- 2.1 IRB banks are required to obtain prior written approval from this Office in terms of the qualitative and quantitative provisions of paragraphs 2.1.1 to 2.1.3 of this directive that cause a rating system change to be regarded as material.
 - 2.1.1 The assessment of materiality shall consist of a qualitative base, which shall include, but not be limited to, the following:
 - 2.1.1.1 the development of a new rating system that is introduced to cater for instances where no rating system existed previously, for example for new jurisdictions and portfolios or mergers and acquisitions;
 - 2.1.1.2 migration between approaches, for example from the foundation IRB approach to the advanced IRB approach;
 - 2.1.1.3 the removal of conservative overlays;¹
 - 2.1.1.4 instances where the Regulations require explicit approval, for example for the assignment of a higher weighting to more recent data in the calculation of the central tendency;
 - 2.1.1.5 the implementation of a new methodology, for example the development of a downturn methodology to replace the United States Federal Reserve's formula in the calculation of the downturn loss given default (LGD);
 - 2.1.1.6 enhancements to the existing methodology, for example the inclusion of multiple defaults in the LGD calculation; and
 - 2.1.1.7 a change in the definition of default.

¹ This refers to cases where conservatism was explicitly incorporated as part of the rating system development due to a lack of reliable historical default and loss experience or weaknesses identified in the rating system.

- 2.1.2 In addition to the above-listed qualitative factors that may trigger changes to be classified as material, the following quantitative factors should also be taken into account when assessing the materiality of the proposed change:
- 2.1.2.1 a rating system change that results in an absolute² change equal to or greater than 1 per cent of the total banking group risk-weighted assets (RWA) relating to credit risk;
 - 2.1.2.2 a rating system change that results in an absolute change equal to or greater than 10 per cent of the RWA relating to credit risk, for the portfolio(s) covered by the relevant rating system;
 - 2.1.2.3 the quantitative measure defined in 2.1.2.1 and 2.1.2.2 above shall exclude all rating system changes where the absolute RWA change is less than R500 million; and
 - 2.1.2.4 the quantitative measures stipulated in 2.1.2.1 to 2.1.2.3 above shall be evaluated based on a comparison between the current rating system and the proposed rating system. Should the changes to the rating system apply to more than one risk parameter (that is, for probability of default (PD), LGD and/or exposure at default (EAD)), the evaluation shall be performed individually for each risk parameter to ascertain whether the proposed change is material.
- 2.1.3 In the event that an IRB bank introduces changes that affect multiple portfolios, for example a change to the definition of default, instead of submitting separate applications for each affected portfolio, the IRB bank shall be required to submit a document outlining the proposed changes as well as any other relevant information related to the proposed changes. The information shall include, but not be limited to, the impact on parameter estimates and regulatory capital for each of the proposed changes for all affected portfolios.
- 2.1.4 This Office will assess the changes deemed to be material and will notify the IRB bank of any objections and/or conditions it may have regarding their implementation.
- 2.2 Non-material rating system changes
- 2.2.1 An IRB bank may implement changes deemed to be non-material (not considered material as per the qualitative and quantitative provisions of paragraphs 2.1.1 to 2.1.2 above) once this Office has received and acknowledged the completed Annexure A. However, to afford this Office the opportunity to raise any objections to and/or to impose any conditions on the implementation of the proposed non-material changes, IRB banks are required to submit for regulatory reporting purposes the information outlined in Annexure A at least ten working days prior to the reporting month in which the proposed changes will be implemented. In the event that this Office does not raise any objections and following acknowledgement of the proposed changes, the IRB bank may implement the proposed changes.³

² An increase or a decrease.

³ The ten-day period will commence only after receipt of the notification.

2.3 Communication

- 2.3.1 All proposed changes, irrespective of whether they are material or non-material, shall be communicated in writing to this Office in the format specified in Annexure A, and shall be signed by the Chief Risk Officer of the IRB bank.
- 2.3.2 For changes that are deemed to be material, the additional information as specified in item 12 of Table 1 of Annexure A shall be appended to the completed Annexure A and communicated to this Office for its consideration and approval prior to implementation.
- 2.3.3 In order to ensure a consistent application of the definition of the materiality of rating system changes by all IRB banks, banks are required to have in place a duly documented communication policy with this Office, approved by the relevant designated committee.⁴ The communication policy shall, among other things, outline the criteria used by each IRB bank for classifying all rating system changes as either material or non-material.
- 2.3.4 To enable this Office to set aside sufficient time to duly consider and review all material changes and any other material aspects related to credit risk rating systems, the aforementioned communication policy shall also contain the explicit requirement for a biannual written communication update to be submitted to this Office on all material developments in relation to each IRB bank's credit risk rating system. However, IRB banks shall communicate immediately, in writing, any matters that may require the urgent attention of this Office prior to the biannual communication updates. These communication updates shall include, among other things, planned rating system developments, recalibrations, material rating system changes, the Credit Risk Rating System Register (refer to <http://www.reservebank.co.za>> Regulation and supervision> Bank Supervision> International standards> Internal ratings-based approaches: Application process and Self-assessment and long form review templates) and any other noteworthy events with respect to the IRB bank's credit risk rating systems. This should be submitted, at the very least, on a biannual basis by 28 February and 31 August based on information as at 31 December and 30 June respectively.

2.4 Documentation

- 2.4.1 The minimum documentation standards specified in paragraphs 2.4.2 to 2.4.4 below shall apply to all additional information as specified in item 12 of Table 1 of Annexure A, when submitting a material rating system change to this Office.
- 2.4.2 The documentation provided shall allow for a full, independent re-performance of the proposed rating system.

⁴ This refers to the designated committee as defined in regulation 39(7) of the Regulations.

- 2.4.3 All submitted documentation shall be the final, signed-off version thereof, and shall comply with the documentation standards as per regulation 23(11)(b)(v)(l) of the Regulations.
- 2.4.4 In addition to the minimum documentation standards as per regulation 23(11)(b)(v)(l) of the Regulations, this Office also requires that the following information be included in the documentation provided:
- 2.4.4.1 an overview of the portfolio(s) rated by the rating system;
 - 2.4.4.2 examples of the types of transactions rated by the rating system;
 - 2.4.4.3 detailed information regarding data used for development, out-of-sample and out-of-time testing, which should include the observation period, performance period, sample size and stability (the IRB bank shall also demonstrate that the data used in the rating system development are representative of the population of the bank's actual borrowers or facilities);
 - 2.4.4.4 changes made from the current to the proposed rating system with reasoning behind the proposed changes;
 - 2.4.4.5 segmentation applied in the rating system, including the final risk parameter estimates and exposure at default (EAD) per segment;
 - 2.4.4.6 a detailed explanation of the proposed methodology to be adopted;
 - 2.4.4.7 any weaknesses or shortcomings of the proposed rating system;
 - 2.4.4.8 RWA and capital impact, including the most granular impact possible for example the RWA and capital impact per change where there are various rating system changes; and
 - 2.4.4.9 in cases where human judgement was used, the IRB bank should clearly document the process for determining and reviewing estimates, which should include all relevant and material information considered other than by the rating system as well as the parties involved.

3. Acknowledgement of receipt

- 3.1 Two additional copies of this directive are enclosed for the use of your institution's independent auditors. The attached acknowledgement of receipt, duly completed and signed by both the chief executive officer of the institution and the said auditors, should be returned to this Office at the earliest convenience of the aforementioned signatories.



René van Wyk
Registrar of Banks

The previous directive issued was Directive 1/2014, dated 09 June 2014.

Application/notification of changes to the internal ratings systems

Table 1: Summary information

1.	Bank name	
2.	Submission date	
3.	Material or non-material change	
4.	Rating system	Name: Unique number: Type (PD, LGD, EAD):
5.	Reason(s) for change(s), particularly if the current rating system has a specific deficiency as opposed to requiring refinement	
6.	Brief description of changes	
7.	Portfolio(s)/business units covered by the rating system	
8.	Regulatory asset class(es) affected by the rating system	Complete the following measures for each regulatory asset class affected by the rating system ⁵ : Regulatory asset class Exposure Rxxx (date) Current EAD Rxxx (date) Current RWA Rxxx (date)
9.	Qualitative factors considered in assessment of materiality	
10.	Relevant committee approval	Committee: Date:
11.	Proposed implementation date for rating system changes	
12.	Supporting documentation for material changes ⁶	Development documentation Validation documentation Documentation presented to approval committee Approval committee minutes ⁷ Other supplementary material
13.	Proposed rating system complies with the minimum requirements as specified in regulation 23(13)(b) of the Regulations	Yes/no
14.	If stated No in item 13: detail the reasons for non-compliance, including actions to address non-compliance	

⁵ Measures should be provided for the total exposure (of the relevant regulatory asset class(es), i.e. Basel II asset class(es)) rated by the rating system (i.e. performing and non-performing loans combined)

⁶ As a minimum, the development documentation provided shall allow for a full, independent re-performance of the proposed rating system. Furthermore, all submitted documentation shall be the final, signed-off version thereof.

⁷ In the event that the approval committee minutes are not yet available, the IRB bank can submit a summary of the discussions and action items from the meeting(s). This summary must be signed by at least one permanent member of the committee.

Table 2: Portfolio and RWA impact information

The measures in the table below should be provided for both the current rating system and the proposed rating system for the portfolio to which they are applied. The measures for the proposed performing loans and non-performing loans (NPLs) should be provided for each risk parameter individually, i.e. probability of default (PD), loss given default (LGD) or exposure at default (EAD), should the changes to the rating system apply to more than one risk parameter.

As at	Exposure (R'm)	EAD (R'm)	EAD- weighted PD (%)	EAD- weighted LGD (%)	RWA (R'm) ⁸	Expected loss (R'm) ⁹	Eligible provisions (R'm)	Capital requirement (R'm) ¹⁰
Current total banking group relating to credit								
Current performing loans								
Current NPLs ¹¹								
Current performing loans (rated sample) ¹²								
Current NPLs (rated sample)								
Proposed performing loans (rated sample if applicable)								
Proposed NPLs (rated sample if applicable)								
Difference ¹³								

⁸ Please note that the scalar of 1,06 should be included in the RWA.

⁹ Expected loss = PD x LGD x EAD for performing loans and best estimate of expected loss (BEEL) for non-performing loans

¹⁰ For the purpose of this directive, this is to be calculated as (RWA x total bank specific minimum required capital percentage) + expected losses – eligible provisions

¹¹ Exposures that are in default as defined in regulation 67 of the Regulations

¹² For rating systems where the capital requirements can only be calculated on a sample of the portfolio covered by the rating system

¹³ The difference shall be calculated between the current and the proposed rating system for the total portfolio.

Table 3: Significance measures

As at:	Change as percentage information			
	EAD (%)	RWA (%)	Expected loss (%)	Capital requirement (%)
Change as a percentage of total banking group relating to credit				
Change as a percentage of the portfolio(s) covered by the rating system				